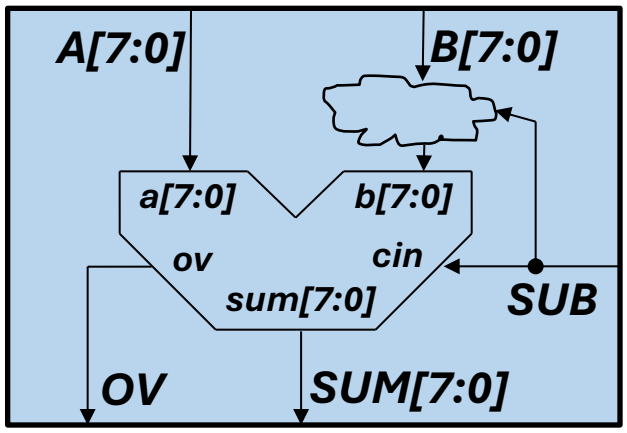
Exercise 04 Arithmetic Block Test & Debug

Create a testbench to test a provided 8-bit arithmetic block **arith.sv**

- This arithmetic unit can perform addition or subtraction.

  SUM = A + B if SUB is low
  SUM = A – B if SUB is high

- It also outputs an overflow signal if the result is either too positive or too negative to represent as an 8-bit signed number.



arith.sv

- Think about what testcases you need to fully test the functionality of this block.
  - What (and how many) input stimuli do I need to verify that addition, subtraction, and overflow work under all conditions?
- **arith** contains a **bug** in the **ov** logic. See if your testbench can find it.
- Fix the bug and in the comments, describe your testcase that caught it.

→ **Submit arith_tb.sv and arith.sv** (fixing the bug in the **ov** logic)